

Program: Diploma in Computer Application and Business Management	
Course Code: 6252C	Course Title: Production and Operations Management
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To provide a thorough knowledge in Production/Operation Management and thus to understand the ways of improving productivity and operational efficiency.

Course Prerequisites: Nil

Course Outcomes:

On completion of the course, students will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the concept of production/operations management, plant layout and process planning	15	Understanding
CO2	Explain Production planning and control, sales forecasting and quality control	14	Understanding
CO3	Summarize Modern production management tools and apply the knowledge of Purchase procedure	15	Applying
CO4	Use Network analysis for project Management	14	Applying
	Series Exam	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1					3		
CO2						3	
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the concept of production/operations management, plant layout and process planning.		
M1.01	Define Production/operations management	1	Understanding
M1.02	Differentiate between Production/operations management	2	Understanding
M1.03	Describe the functions of Production/operations management	2	Understanding
M1.04	Explain the different types of production systems	2	Understanding
M1.05	Explain plant layout	2	Understanding
M1.06	Recall plant layout and its different types	2	Understanding
M1.07	Recall process planning	1	Understanding
M1.08	Demonstrate process charts	1	Understanding
M1.09	Describe different types of process charts	2	Understanding
Contents: Production/operation management – meaning – Differences - functions - types of production system – plant layout – types of plant layout - Process planning – meaning – process chart – types of process charts.			
CO2	Explain Production planning and control, sales forecasting and quality control.		
M2.01	Explain production planning and control	2	Understanding
M2.02	Recall Sales forecasting	1	Understanding
M2.03	Outline the different techniques of sales forecasting	2	Understanding

M2.04	Explain quality control	1	Understanding
M2.05	State the objectives of quality control	1	Understanding
M2.06	Explain Statistical quality control	2	Understanding
M2.07	Recall capacity	1	Understanding
M2.08	State the concept of capacity planning	2	Understanding
M2.09	Explain aggregate planning	2	Understanding
	Series Test-I	1	
Contents: Production planning and control. – Sales forecasting – techniques of sales forecasting - Quality control – objectives - Statistical quality control – capacity – meaning - capacity planning - Aggregate planning.			
CO3	Summarize Modern production management tools and apply the knowledge of Purchase procedure.		
M3.01	Explain procurement	1	Understanding
M3.02	Explain Procurement procedure	2	Understanding
M3.03	Describe various methods of purchasing – case study	2	Applying
M3.04	Explain centralized and decentralized purchasing	2	Understanding
M3.05	Examine the five ‘R’ principles of purchasing	2	Understanding
M3.06	Explain value analysis	1	Understanding
M3.07	Explain codification	1	Understanding
M3.08	Explain simplification	1	Understanding
M3.09	Explain Modern production management tools	3	Applying
Contents: Procurement – procedure – purchasing - methods of purchasing - centralized and decentralized purchasing - five R principles - value analysis - codification – simplification - Modern production management tools – Just – In - Time, Total Quality Management - ISO 9000 Series, Kaizen & Business Process Reengineering – case studies.			
CO4	Use Network analysis for project Management.		
M4.01	Describe Network Analysis and state its objectives	1	Understanding

M4.02	Explain Programme Evaluation Review Technique (PERT) with sample problems	3	Applying
M4.03	Explain Critical Path Method with sample problems	3	Applying
M4.04	State the concept of project	1	Understanding
M4.05	Describe the characteristic of projects	1	Understanding
M4.06	Explain project management	1	Understanding
M4.07	Identify the phases of project management	2	Understanding
M4.08	Explain the sources of project ideas	2	Understanding
	Series Test-II	1	
Contents: Network analysis – Objectives – PERT & CPM with sample problems - Project - characteristics of projects - project management - phases of project management - sources of project ideas.			

Text / Reference:

T/R	Book Title/Author
R1	Russell, Roberta S, and Bernard W.Taylor III, Operations Management , Pearson Education, New Delhi 2004.
R2	Aswathappa, Production/Operations Management , Himalaya publishing
R3	Chary, S,N., Production and Operations Management , Tata McGraw Hill, New Delhi 1989
R4	Norman Gaither, Production/Operations Management , South-Western Pub, 1998, 8th Edition.

Online Resources:

Sl.No	Website Link
1	https://managementstudyguide.com/portal/courses/categories/98
2	https://www.tutor2u.net/business/topics/production
3	https://www.vskills.in/certification/tutorial/production-and-operations-management-tutorials/
4	https://www.wisdomjobs.com/e-university/production-and-operations-management-tutorial-295/production-management-9437.html
5	https://www.tonex.com/operations-management-tutorial/